higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

COMPUTER SOFTWARE AND HARDWARE
NQF LEVEL 3

26 FEBRUARY 2019

This marking guideline consists of 7 pages.

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SECTION A

QUESTION 1

1.1 True
1.2 True
1.3 True
1.4 True
1.5 True
1.6 False
1.7 False
1.8 True
1.9 True
1.10 False

(10 × 1) **[10]**

QUESTION 2

2.1 C
2.2 I
2.3 A
2.4 J
2.5 K
2.6 B
2.7 F
2.8 H
2.9 E
2.10 D

(10 × 1) **[10]**

QUESTION 3

3.1 A
3.2 D
3.3 B
3.4 D
3.5 D
3.6 C
3.7 B
3.8 A
3.9 C
3.10 A

(10 × 1) **[10]**

TOTAL SECTION A: 30

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SECTION B

QUESTION 4

- Application software✓ is a program used to perform a specific task or is any program or group of programs, that is designed for the end user.✓
- Language translators✓ are programs that convert high-level languages into low-level languages.✓
- Utility programs✓ are specialised system software programs that perform one specific task.✓
- Software tools✓ are programs used to create, debug or maintain other programs✓
- System software✓ is a type of software that manage a computer's resources, scheduling and the monitoring of computer events.✓ (5 × 2) **[10]**

QUESTION 5

5.1
- The number of concurrent users on the system
- The speed rating of the network cards.
- The quality of the cabling or data links
- The speed of devices like routers, hubs and switches
- The speed and capacity of the fileserver
- The speed and processing power of the workstation
- The application being processed
- The corrected functioning of all the devices that provides the communication links for the network

(Or any other applicable answer) (Any 6 × 1) (6)

5.2 5.2.1 Benchmarking – is the process of testing working components by running a computer program that tells you if a component is working as the manufacturer intended it to work.

5.2.2 User-acceptance sign-off – is a process that a computer technician follows when he/she has finished installing a computer system.

(2 × 2) (4)

5.3 5.3.1 Parallel ports – are sockets on a computer's motherboard used to connect a printer or other peripheral devices.
Parallel ports are used as an alternative to the slow-speed serial ports.

5.3.2 Serial ports – are used to connect modems, acquisition terminals or other serial devices via slow-speed serial interface.
Transmits data a bit at a time.

5.3.3 USB ports – are used to connect USB devices and transfer digital data over USB cables.

(3 × 2) (6)

5.4 A hub, also called a network hub, is a common connection point for devices in a network. Hubs are devices commonly used to connect segments of a LAN. (2)

5.5 A hub contains multiple ports.✓ When a packet arrives at one port, it is copied to the other ports so that all segments of the LAN can see all packets.✓ (2)
[20]

QUESTION 6

- 1GHz or higher or faster Processor
- 1GB RAM (32-bit) or 2GB RAM (64-bit)
- 16 GB Hard drive or higher
- Graphics card or device

[4]

QUESTION 7

7.1
- You should consider the checklist to ensure that you meet the user's requirements and it will minimise the time you spend on the installation.
- You should consider the following important factors:
- What software the client (Labs or Staff) wants installed.
- Is it easily available in the country or must be brought from overseas.
- If the software is appropriate for the user
- The user should have a specification of their software requirements
- To establish the feasibility of the specification, it is a good idea to ask the user about their requirements
- Whether the PC meets the minimum requirements of the software
- Check that the hardware meets the recommended specifications
- Software will run if the hardware meets the specification, but it will not run well or quickly.
- All software is supplied with minimum and recommended specifications.
- Whether the software needs customisation or special setting to meet the user requirements. (Or any other applicable answer) (Any 10 × 1) (10)

7.2
- A driver interacts with the device through a communication subsystem to which the hardware connects.
- The operating system or other computer program calls a function in the driver to execute commands on the device.
- Device drivers act as a translator between the operating system and the hardware device.
- There are drivers for almost every computer device and peripheral such as mouse, keyboard, hard disk drive, network interface cards, floppy disk drives, network adapters, SCSI adapters, sound cards, printers and modems.
- The computer's operating system cannot interact with a device if its driver is not properly installed and configured on the computer. (Any 3 × 2) (6)

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7.3
- It is the heart of the computer and this is where all the computing or processing takes place.
- It performs all arithmetic and logic operations and controls all other functions on the PC
- The CPU contains the instruction set, arithmetic logic unit, data paths, registers and cache.
- No program will load, no calculations will be made, no messages displayed without the approval of the CPU
- The CPU is the component in charge of everything that happens on the PC.
- When the PC is switched on, and the power flows through the circuits, the first thing that the CPU does is to check the contents of the CMOS, it does this so that it knows what the system configuration is and what devices are attached.
- The CPU then checks the I/O devices for an operating system such as Windows XP or Linux or the Mac OS.
- The CMOS will have information that tells the CPU which devices to search, and in what sequence, when looking for the operating system.

(Any 4 × 1) (4)

[20]

QUESTION 8

8.1 8.1.1
- Time✓ – Determine whether the installation can be completed within the prescribed timeframe or not.✓
- Budget✓ – Determine whether the installation can be completed with the allocated budget or not.✓
- Resource availability✓ – Check if all required resources are available or not. If not, find out if they can be ordered and delivered within the prescribed time frame.✓
- Software licenses✓ – Determine if all required software is available and fully licensed.✓
- Current workload✓ – Determine the amount of other work that still needs to be done.✓

(Or any other applicable answer) (5 × 2) (10)

8.1.2
- Fewer user complaints
- Less customer frustration
- Problems with the computer are picked up before the computer is released to its new owner
- Less return jobs (Any 2 × 1) (2)

8.1.3 Sanity testing is a very brief run-through of the functionality✓ of a computer system to ensure that the system works as expected. It is often done before a more exhaustive round of testing✓. (2)

8.2
- Receive the particulars of the users who sent the notifications of computer problems.
- Record the type of problems using good communication techniques. (3)

- Prioritise the problems according to the order of preference.

8.3
- Define the problem.
- Collect data and evidence.
- Identify the issue.
- Find the root cause.
- Develop recommendations for possible solutions.
- Implement the selected solution.
- Monitor the solution to ensure its effectiveness. (Any 4 × 1) (4)

[21]

QUESTION 9

9.1
- Hardware incompatibility.
- Not wearing antistatic wrist strap.
- Improper handling of the motherboard during installation.
- No hardware device drivers.
- Damaged hardware.
- Shortage of installation tool kit.
- Insufficient installation knowledge.
- Shortage of other devices not delivered in time. (Any 5 × 1) (5)

9.2
- Scope of testing
- Schedule
- Test deliverables
- Aspects not tested
- Release criteria
- Risk contingencies
- Compiled documents for the report (Any 5 × 1) (5)

[10]

QUESTION 10

10.1
- Monitor must be cleaned using the correct material such as glass cleaner and soft cloth.
- Switching of the computer if not in use.
- Computer must be kept out of direct sunlight.
- Handle hardware components properly to avoid physical damage.
- Always wear an antistatic wrist strap to eliminate static charges.
- Avoid eating and drinking while working on a computer.
- Install the computer at a humidity-controlled environment.

(Or any other applicable answer) (Any 5 × 1) (5)

10.2
- Moisture
- Humidity
- Dust
- Heat
- Magnetic field

- Electrical surges
- Electrostatic discharge (Any 6 × 1) (6)

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10.3
- Troubleshooting refers to a way of solving problems that are based on the process of elimination of all the possible causes of a problem.
- A diagnosis is a process of working out what is wrong, so that it can be rectified. (2 × 2) (4)

10.4
- User manuals
- Information logs
- Manufacturer's website
- Experts
- Internet discussion forums (Or any other applicable answer) (5)

[20]

QUESTION 11

11.1
- Arrange suitable packaging.
- Arrange time to be at the destination office or building so that someone will be there to receive.
- Get clearance from your manager.
- Write down serial numbers of components in case of damage or theft.
- Make sure all the components have been packed.
- Make sure the boxes or packaging is sealed.
- Organise proper transport such as a trolley to transport the boxes.
- Store the boxes in an area that is free from water, humidity and sunlight.

(Or any other applicable answer) (Any 5 × 1) (5)

11.2
- Internet
- Software manufacturers
- Hardware Manufacturers
- Computer shops (Any 3 × 1) (3)

11.3
- Ensure the mouse is within the range of the receiver.
- Replace the batteries in the wireless mouse.
- Refer to the documentation that came with the mouse

(Or any other applicable answer) (3)

11.4
- Run the disk clean-up program to delete unwanted files.
- Run the disk defragmenter program to group fragmented files together on the hard disk drive.
- Remove unused programs.
- Check for hard disk drive errors.
- Close all open programs and try the desired task again.

(Or any other applicable answer) (Any 4 × 1) (4)

[15]

TOTAL SECTION B: 120
GRAND TOTAL: 150

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